

SOAP Shape Tool – Technical Documentation

1. Design Decisions

The system uses Spring Web Services with a contract-first approach. The XML schema defines structure before implementation, ensuring strong validation, loose coupling, and interoperability.

2. Schema Structure

The schema (shapetool.xsd) defines request and response elements for each shape. Each operation has separate request and response elements to maintain clarity and structure.

3. Namespace Usage

Namespace: `http://example.com/shapetool`

Used across XSD, Java endpoint, and SOAP requests to ensure consistency and avoid conflicts.

4. Validation Strategy

Two levels:

- Schema validation: Ensures correct structure and data types
- Application validation: Ensures logical correctness (e.g., positive values)

5. Service Operation Design

Each shape has its own SOAP operation. Methods are stateless and follow a consistent request-response pattern.

6. Error Handling

Invalid inputs throw runtime exceptions, which are converted into SOAP Fault responses. This ensures structured error communication.

7. WSDL Generation

The WSDL is auto-generated using Spring configuration and is accessible via `/ws/shapetool.wsdl`.

Summary

The system demonstrates contract-first SOAP design, strict validation, modular architecture, and structured error handling.